WHAT EXISTS · WHAT WE ADAPTED

Phones and VPNs already move calls between networks. None checks a network before using it.

EXISTING MECHANISM	DOES WELL	MISSES	IN OUR RESULTS · SIMULATED
The phone's own switch Wi-Fi → mobile when Wi-Fi is poor	Built in, no setup	Moves after the call suffers; joins any saved name, even a copy	**6.56** s frozen leaving a building
Multipath TCP RFC 8684	Keeps a TCP connection over Wi-Fi and mobile	TCP only, not real-time voice; each app must support it	**1.6** s median reconnect, from iPhone measurements
Roaming VPN tunnel WireGuard, NDSS 2017	The call survives a new address, sealed	Doesn't know when to move, or which networks are safe	**8.29** switches per café call
Operator steering 3GPP ATSSS · TS 23.501	The operator's core splits traffic over 5G and Wi-Fi	Needs the operator's support; no help on café, venue or satellite networks	Not modelled
Enterprise roaming IEEE 802.11r	Moves between access points in under 50 ms	Only inside one network	Used inside Infinet

1 Check before use

Screen every new network, then probe it in quarantine with a separate key.

2 Move before it breaks

See the Wi-Fi edge on the coverage map; send on both networks for 2 s.

3 Seal everything

One tunnel on every network: even a perfect copy sees only ciphertext.

4 Patient on satellite

Adaptive timeouts (RFC 6298) and the learned LEO pattern, not a fixed timeout.

REFERENCES J. A. Donenfeld, “WireGuard: Next Generation Kernel Network Tunnel”, NDSS 2017 · 3GPP TS 23.501, Access Traffic Steering, Switching and Splitting (ATSSS) · IETF RFC 8684, Multipath TCP · IETF RFC 6298, retransmission timer

WHAT WE'D HAND AN OPERATOR OR A PHONE MAKER

Seven rules keep a moving call alive and private.

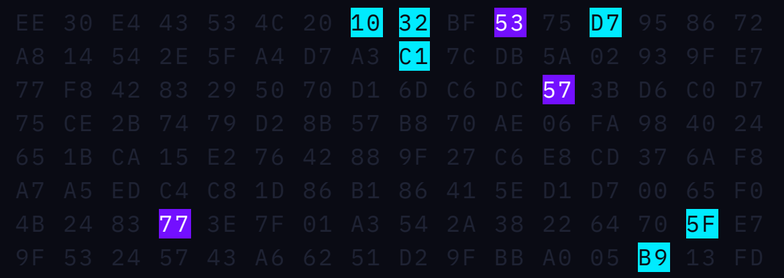
	RULE	VALUE WE USE	EVIDENCE ☐ SIMULATED · 300 RUNS PER SCENE
1	**Hold the voice target**	One-way delay ≤ 150 ms (ITU-T G.114); quality from the E-model (ITU-T G.107); conceal up to 2 lost frames	**4.34** vs 3.72 call quality (MOS) leaving a building · Infinet vs today's phone
2	**Prepare early, switch late**	Check the next network 5 s before the Wi-Fi edge; switch with under 1.5 s left; Wi-Fi usable above −75 dBm	**3.4 s early** average lead before the Wi-Fi faded · 0 s frozen
3	**Make before break**	Send on two tunnels for 2 s around a switch, only where the delay meets the playout deadline	**+2–4.4 %** extra data, by scene: the cost of the second copy
4	**Never use an unchecked network**	Screen, then probe in quarantine; timeout max(1 s, 3 × round trip); one flag caps at Tunnel only, two serious flags block	**0 frames** over a fake, unchecked or Blocked network in 5,000 runs
5	**Seal everything, fail closed**	Every frame in the tunnel on every network; anchor down → nothing in the clear, call ends after 8 s	**Ciphertext only** all that a perfect copy of the café Wi-Fi sees
6	**Be patient on satellite**	Adaptive timeouts (RFC 6298); learn the 15 s LEO pattern and protect 300 ms before each spike	**0** vs 10.4 false “link dead” calls per run · Infinet vs tunnel, no check
7	**Don't flap**	Wait 4 s before returning to a Wi-Fi, doubling to 60 s; Wi-Fi first only when checked and unflagged	**2.75** vs 8.29 switches per café call · Infinet vs tunnel, no check

Settings marked as assumptions in our settings file are to be tuned with field data. In the desert the call still freezes about 5 s during the satellite failover; the difference is that it is never dropped.

DECLARATION

We used AI, and we declare it.

Claude (Anthropic), through Claude Code, was our coding and writing assistant. The idea, the decisions and the survey are ours.



THE ASSISTANT

What it did

- Wrote most of the simulation code, tests and tools
- Built the pipeline that renders our film
- Drafted documents from our notes and decisions
- Most of our commits are marked as co-written with it

THE TEAM

What we did

- The idea and the problem framing
- The venue survey, walking the hall with our own phones
- The scenarios, the baselines and every design decision, logged with its reason
- Reviewed and tested what it wrote

HONESTY

How we checked it

- Every number traces to our settings file: cited, or marked as an assumption (**96** of **163**)
- **68** automated tests
- **300** seeded runs per scene, not hand-picked; demo runs chosen near the median

ASK US

Each of us can explain the tunnel, the trust check and the results in our own words, without the tool.